Pipeline ML On-Chain v2
Token UNI (Uniswap) - Real Data

Generated: 19/04/2026 17:21

Frequencies: 1h, 4h, daily

Models: XGBoost Baseline, XGBoost Tuned, LSTM

Data: Dune Analytics + The Graph + Binance + Fear&Greed

1. Executive Summary

1.1 v1 (Synthetic Data) vs v2 (Real Data)

V1 pipeline used synthetic/simulated Dune data for development and testing.
V2 pipeline uses real data from multiple on-chain sources:
- Dune Analytics: real UNI holder counts and whale metrics
- The Graph: real Uniswap V3 protocol metrics (volumeUSD, TVL, priceUSD, feesUSD)
- Binance API: real hourly OHLCV data (1h, 4h, daily resample)
- Alternative.me: Fear & Greed Index (daily sentiment signal)

Key improvements in v2:
* Multi-frequency pipeline: 1h, 4h, and daily datasets
* Transaction cost modeling: 0.1% per half-turn applied on signal changes
* Max drawdown metric added to all model evaluations
* Expanded feature set: fear_greed_value, volumeUSD lags, price_divergence
* min_child_weight added to XGBoost hyperparameter search grid
* Configurable LSTM lookback per frequency

1.2 Pipeline Overview

Freq	Rows	Features	Best Model
1h	43789	180	XGB_baseline: acc=56.83% sharpe=-12.60
4h	10950	180	XGB_tuned: acc=61.32% sharpe=-3.24
daily	1825	180	XGB_tuned: acc=50.96% sharpe=-1.42

2. Data Sources Description

2.1 Binance Hourly OHLCV

Source: Binance REST API (no API key required)
Endpoint: GET https://api.binance.com/api/v3/klines
Symbol: UNIUSDT, Interval: 1h
Coverage: last 5 years (1825 days), paginated at 1000 candles/request
Columns: datetime, open, high, low, close, volume
Cache: data/cache/binance_uniusdt_1h_1825d.parquet (1-day TTL)
Resampled to 4h and daily for multi-frequency pipeline

2.2 Dune Analytics Holders

Source: data/dune_uni_holders.csv (real Dune Analytics export)
Content: daily UNI holder counts, whale metrics, concentration indexes
Key columns: date, number_of_holders, whale_count, whale_balance_pct, whale_balance_total, herfindahl_index
Derived: holders_growth (pct change), whale_delta_pct (pct change)

2.3 The Graph Protocol Metrics

Source: data/thegraph_uni_protocol.csv (real The Graph export)
Content: daily Uniswap V3 protocol metrics for UNI token
Key columns: date, volumeUSD, totalValueLockedUSD, priceUSD, feesUSD
Derived: volume_tvl_ratio, fee_revenue, price_divergence

2.4 Fear and Greed Index

Source: https://api.alternative.me/fng/?limit=2000&format=json
Content: daily crypto market sentiment index (0-100)
Encoding: Extreme Fear=0, Fear=1, Neutral=2, Greed=3, Extreme Greed=4
Key columns: date, fear_greed_value, fear_greed_class
Cache: data/cache/fear_greed.parquet (1-day TTL)

3. Dataset Statistics per Frequency

3.1 Frequency: 1h

Metric	Value	Detail
Date range	2021-04-20 21:00	2026-04-19 19:00
Total rows	43789	
Train rows	35031	80%
Test rows	8758	20%
Features	180	
Min price (USD)	3.0170	
Max price (USD)	44.6440	
Avg price (USD)	9.6903	
Avg holders	1255	
Avg Fear&Greed	45.8	0-100

3.2 Frequency: 4h

Metric	Value	Detail
Date range	2021-04-20 20:00	2026-04-19 16:00
Total rows	10950	
Train rows	8760	80%
Test rows	2190	20%
Features	180	
Min price (USD)	3.0180	
Max price (USD)	44.4910	
Avg price (USD)	9.6930	
Avg holders	1256	
Avg Fear&Greed	45.8	0-100

3.3 Frequency: daily

Metric	Value	Detail
Date range	2021-04-20 00:00	2026-04-18 00:00
Total rows	1825	
Train rows	1460	80%
Test rows	365	20%
Features	180	
Min price (USD)	3.0180	
Max price (USD)	43.1080	
Avg price (USD)	9.6993	
Avg holders	1257	
Avg Fear&Greed	45.8	0-100

4. Features List

Feature categories across all frequencies:

Price features (from Binance):
  log_return, pct_return, price_ma24, price_ma168, volatility_24h,
  high_low_range, volume_ma24

Holder features (from Dune):
  number_of_holders, holders_growth, whale_count, whale_balance_pct,
  whale_balance_total, whale_delta_pct, herfindahl_index

Protocol features (from The Graph):
  volumeUSD, totalValueLockedUSD, priceUSD, feesUSD,
  volume_tvl_ratio, fee_revenue, price_divergence

Sentiment features (from Fear & Greed):
  fear_greed_value, fear_greed_class

Whale signal features (derived):
  whale_accumulation, whale_distribution, whale_confluence,
  sell_pressure, concentration_ratio, whale_dominance

Lag features (frequency-specific periods):
  1h: lags at [1, 4, 24, 168] periods
  4h: lags at [1, 6, 18, 42] periods
  daily: lags at [1, 2, 3, 7] periods

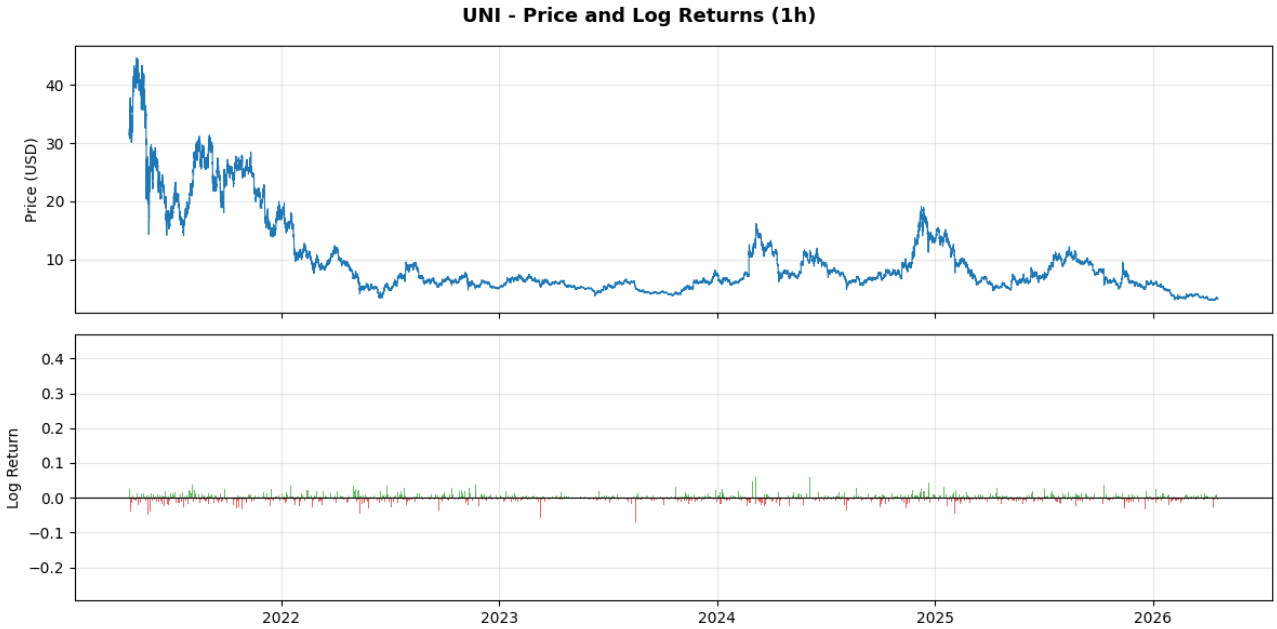
Rolling window features (frequency-specific):
  1h/4h: rolling windows [24, 72, 168]
  daily: rolling windows [7, 14, 30]

Target variables:
  target_1h: next period log_return > 0 (shift -1)
  target_4h: cumulative log_return next 4 periods > 0
  target_24h: cumulative log_return next 24 periods > 0
  (for 4h/daily: shift(-1) equivalents)

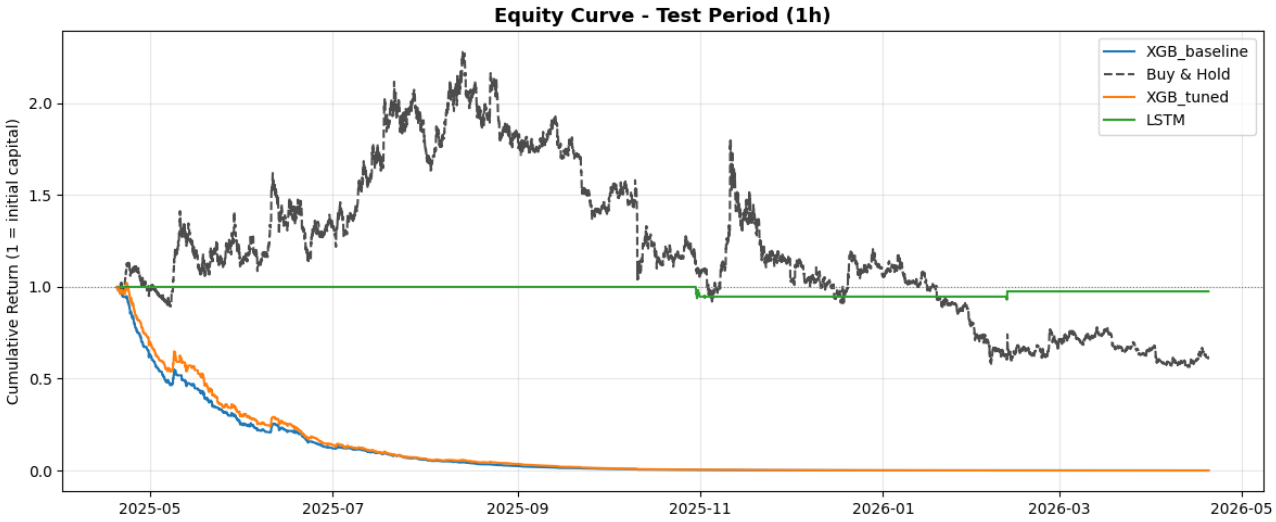
5. Results - Frequency: 1h

5.1 Comparison Table

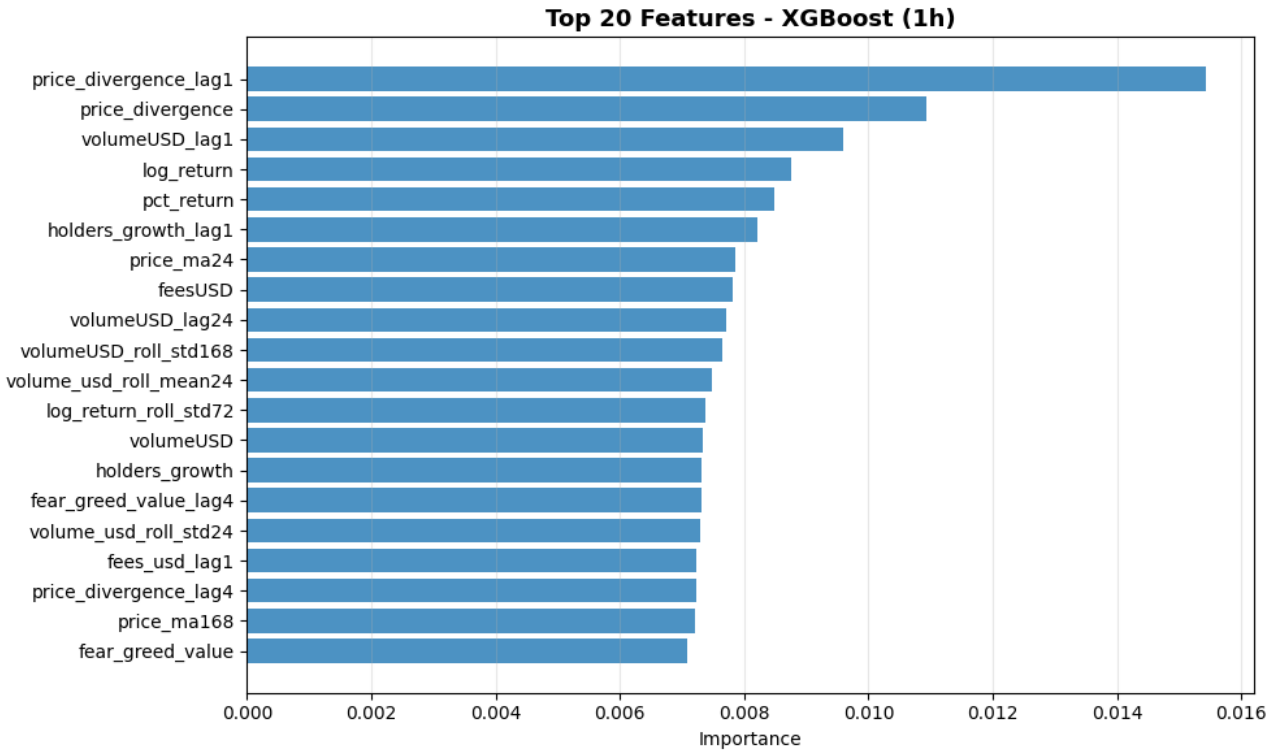
Model	Dir.Acc	ROC-AUC	Sharpe	Return	MaxDD
XGB_baseline	56.83%	0.5883	-12.596	-100.00%	-100.00%
XGB_tuned	56.74%	0.6002	-11.662	-100.00%	-100.00%
LSTM	51.41%	0.5001	-0.296	-2.42%	-6.93%
Buy & Hold	-	-	0.003	-38.45%	-



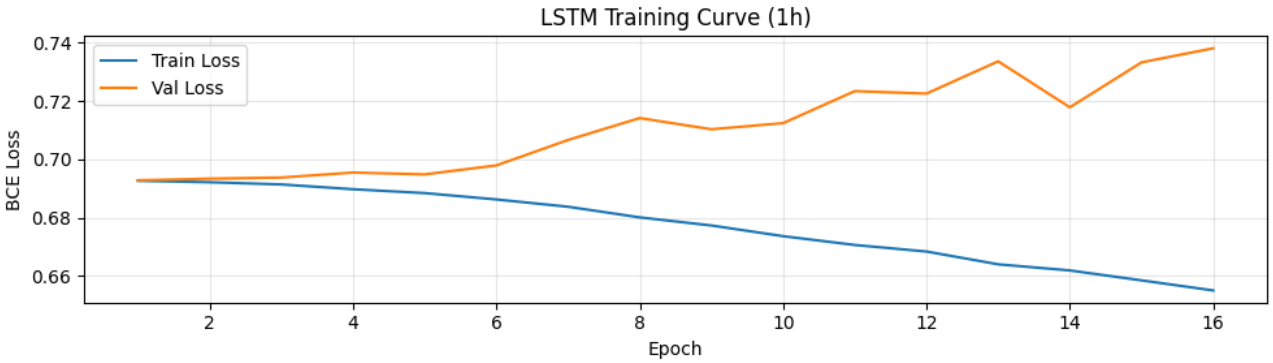
5.2 Equity Curve



5.3 Feature Importance (XGBoost)



5.4 LSTM Training Curve



5.5 Walk-Forward Validation

Fold	Accuracy	ROC-AUC	Train	Test
Fold 1	0.5400	0.5439	7298	7298
Fold 2	0.5038	0.5301	14596	7298
Fold 3	0.5533	0.5793	21894	7298
Fold 4	0.5549	0.5808	29192	7298
Fold 5	0.5626	0.5860	36490	7298
MEAN	0.5429	0.5640	-	-

5. Results - Frequency: 4h

5.1 Comparison Table

Model	Dir.Acc	ROC-AUC	Sharpe	Return	MaxDD
XGB_baseline	58.77%	0.6274	-4.272	-98.50%	-98.62%
XGB_tuned	61.32%	0.6554	-3.241	-96.72%	-96.91%
LSTM	51.33%	0.5001	0.669	24.89%	-16.56%
Buy & Hold	-	-	0.030	-37.95%	-

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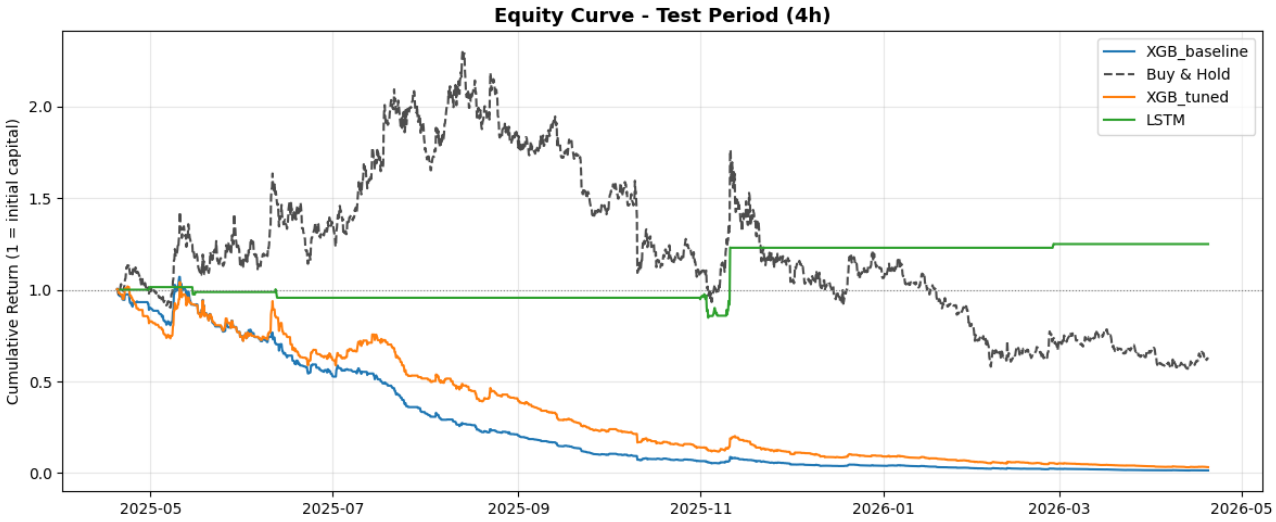

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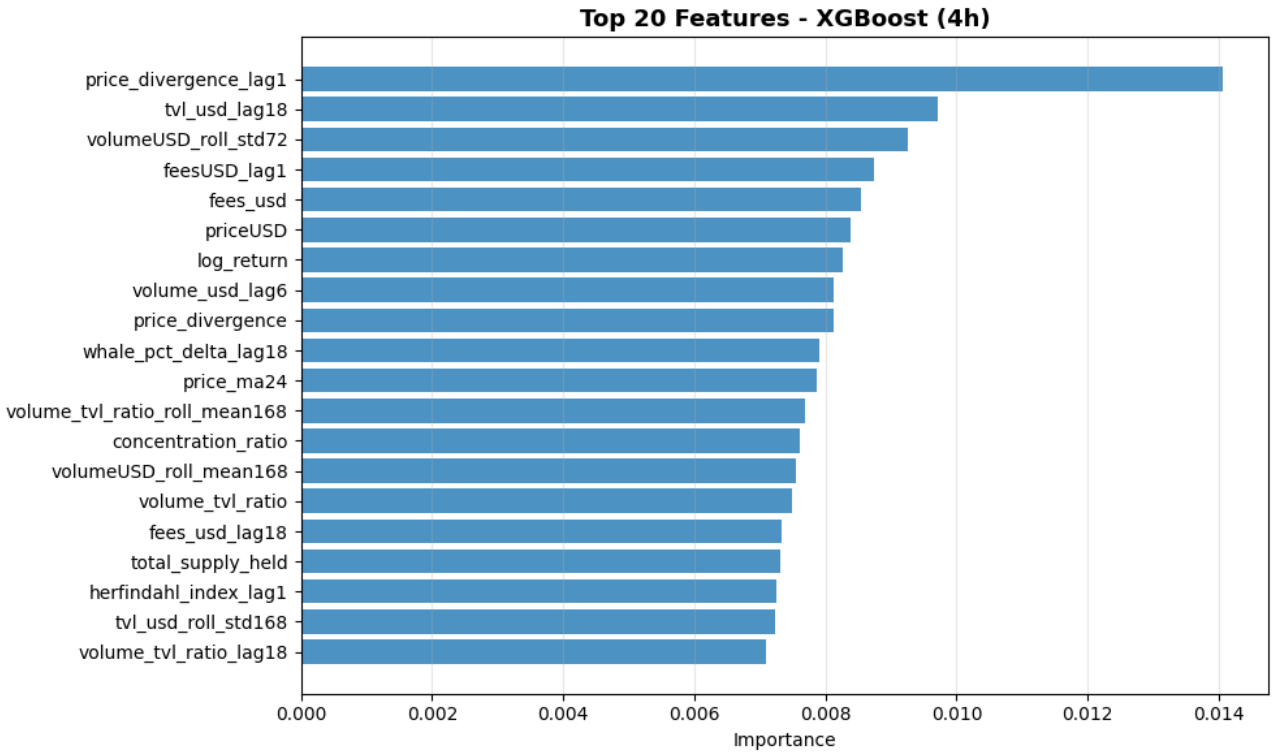

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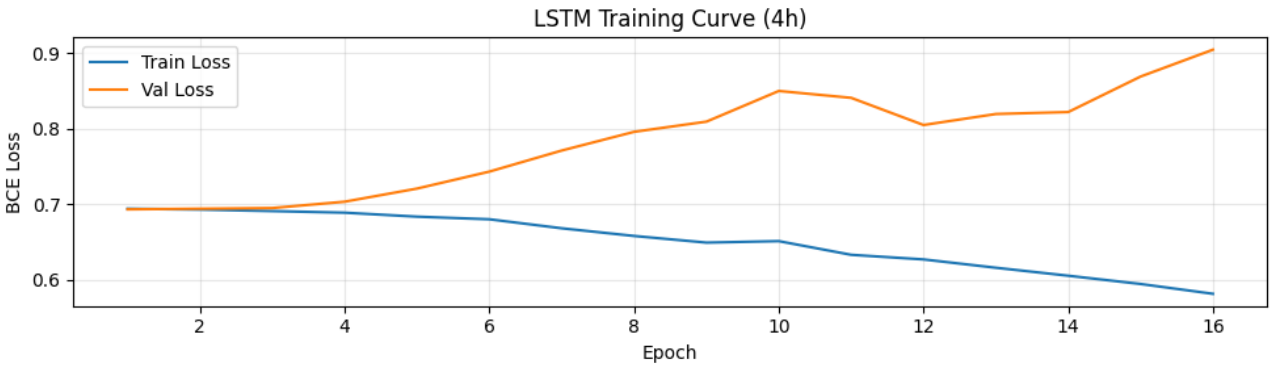


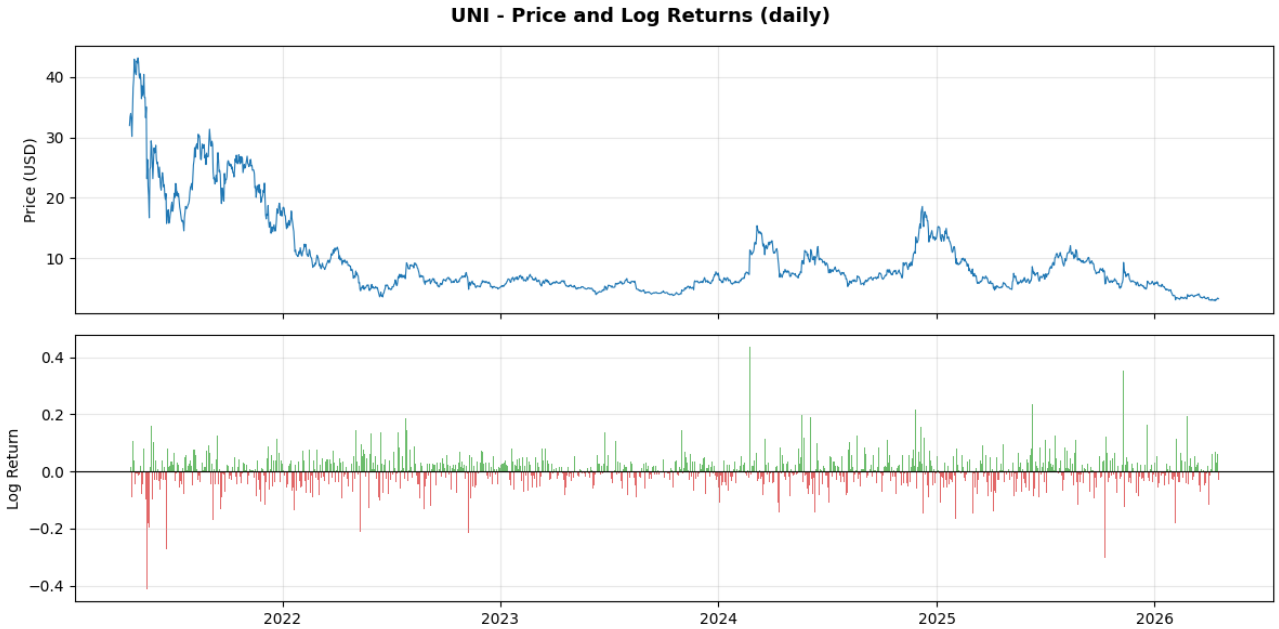
5.5 Walk-Forward Validation

Fold	Accuracy	ROC-AUC	Train	Test
Fold 1	0.5408	0.5607	1825	1825
Fold 2	0.5463	0.5680	3650	1825
Fold 3	0.5732	0.6006	5475	1825
Fold 4	0.5715	0.6066	7300	1825
Fold 5	0.5885	0.6219	9125	1825
MEAN	0.5641	0.5916	-	-

5. Results - Frequency: daily

5.1 Comparison Table

Model	Dir.Acc	ROC-AUC	Sharpe	Return	MaxDD
XGB_baseline	47.95%	0.4723	-8.373	-99.61%	-99.62%
XGB_tuned	50.96%	0.5193	-1.415	-72.61%	-75.57%
LSTM	43.02%	0.4678	0.004	-31.27%	-67.17%
Buy & Hold	-	-	0.098	-35.25%	-



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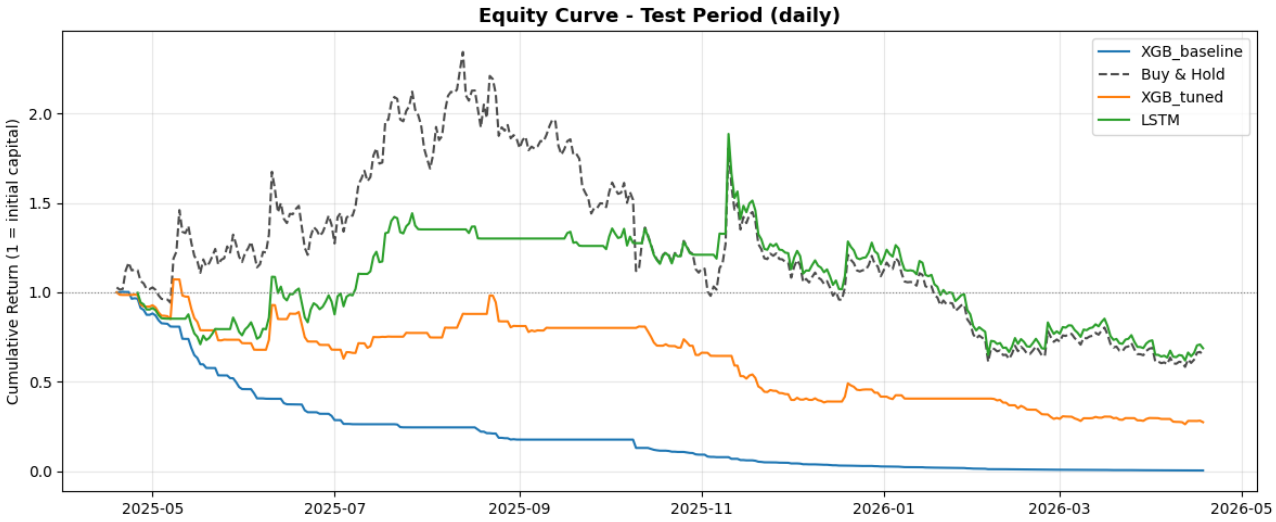

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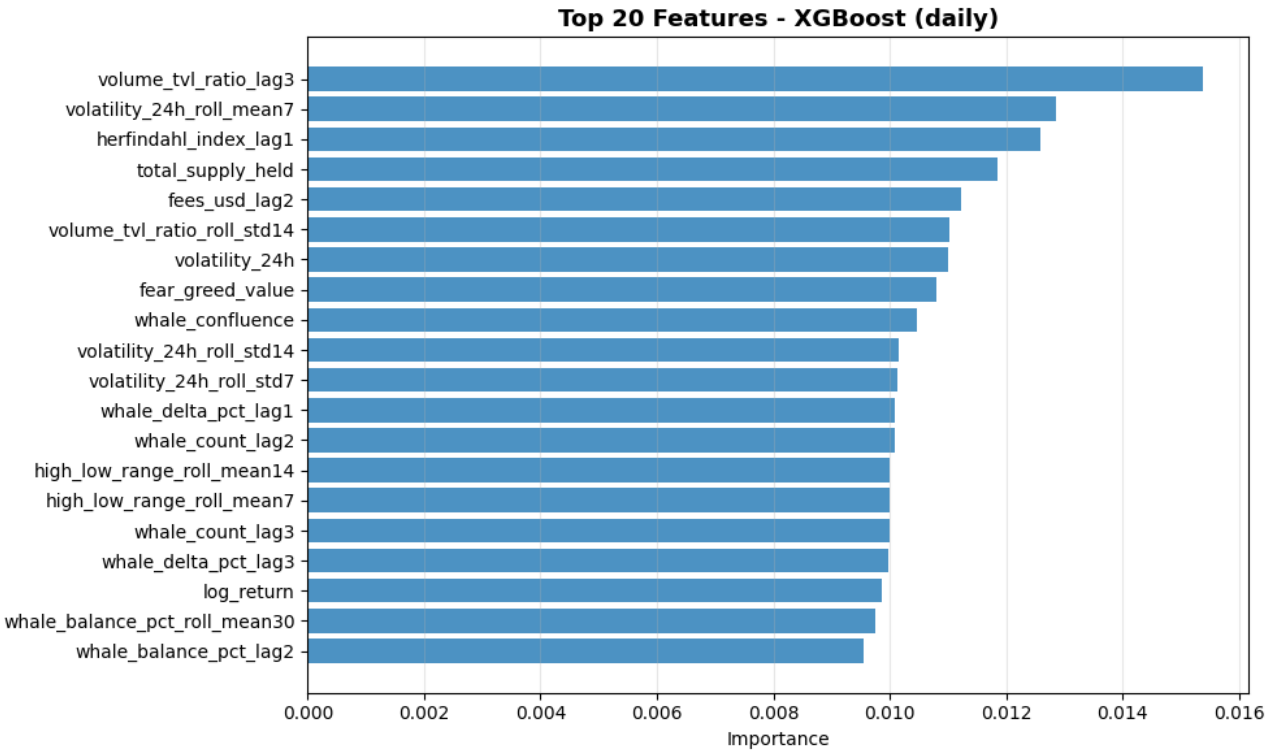

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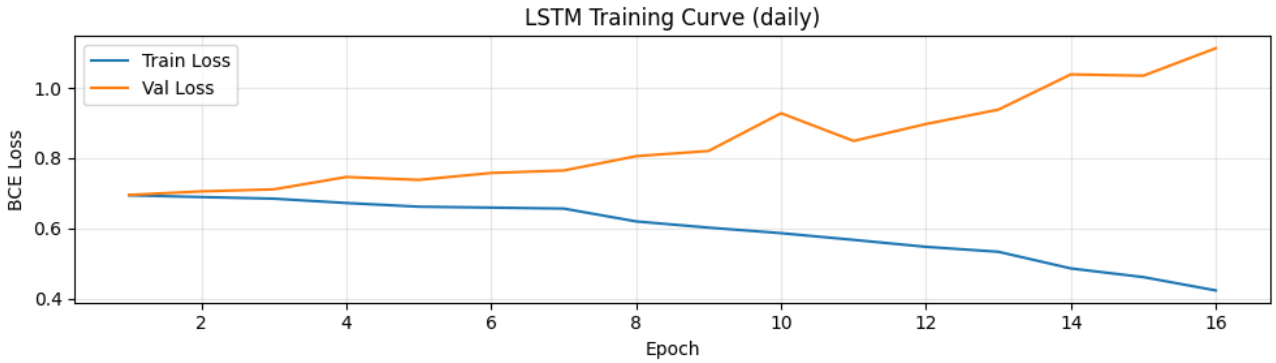
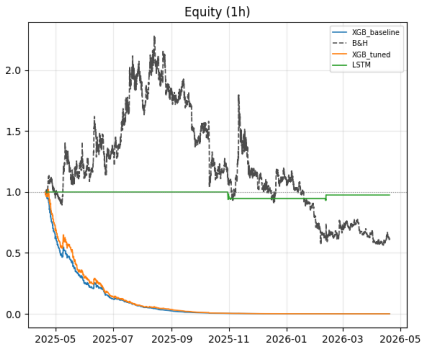


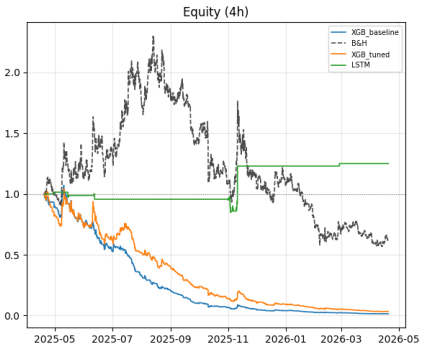
5.5 Walk-Forward Validation

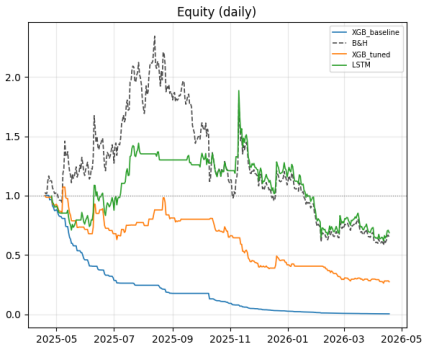
Fold	Accuracy	ROC-AUC	Train	Test
Fold 1	0.5132	0.5659	304	304
Fold 2	0.4408	0.4381	608	304
Fold 3	0.5230	0.5194	912	304
Fold 4	0.5329	0.5509	1216	304
Fold 5	0.5559	0.4669	1520	304
MEAN	0.5132	0.5082	-	-

6. Master Table: All Models, All Frequencies, All Metrics

Freq	Model	Dir.Acc	AUC	IC	Sharpe	Return	B&H Ret	MaxDD
1h	XGB_baseline	56.83%	0.5883	0.1529	-12.596	-100.00%	-38.45%	-100.00%
1h	XGB_tuned	56.74%	0.6002	0.1735	-11.662	-100.00%	-38.45%	-100.00%
1h	LSTM	51.41%	0.5001	0.0002	-0.296	-2.42%	-37.50%	-6.93%
4h	XGB_baseline	58.77%	0.6274	0.2207	-4.272	-98.50%	-37.95%	-98.62%
4h	XGB_tuned	61.32%	0.6554	0.2691	-3.241	-96.72%	-37.95%	-96.91%
4h	LSTM	51.33%	0.5001	0.0002	0.669	24.89%	-37.66%	-16.56%
daily	XGB_baseline	47.95%	0.4723	-0.0492	-8.373	-99.61%	-35.25%	-99.62%
daily	XGB_tuned	50.96%	0.5193	0.0336	-1.415	-72.61%	-35.25%	-75.57%
daily	LSTM	43.02%	0.4678	-0.0555	0.004	-31.27%	-42.52%	-67.17%







7. Transaction Costs Analysis

Transaction cost model used in v2:

Cost per half-turn: 0.001 (0.1%)
A complete round-trip (buy + sell) costs: 0.002 (0.2%)

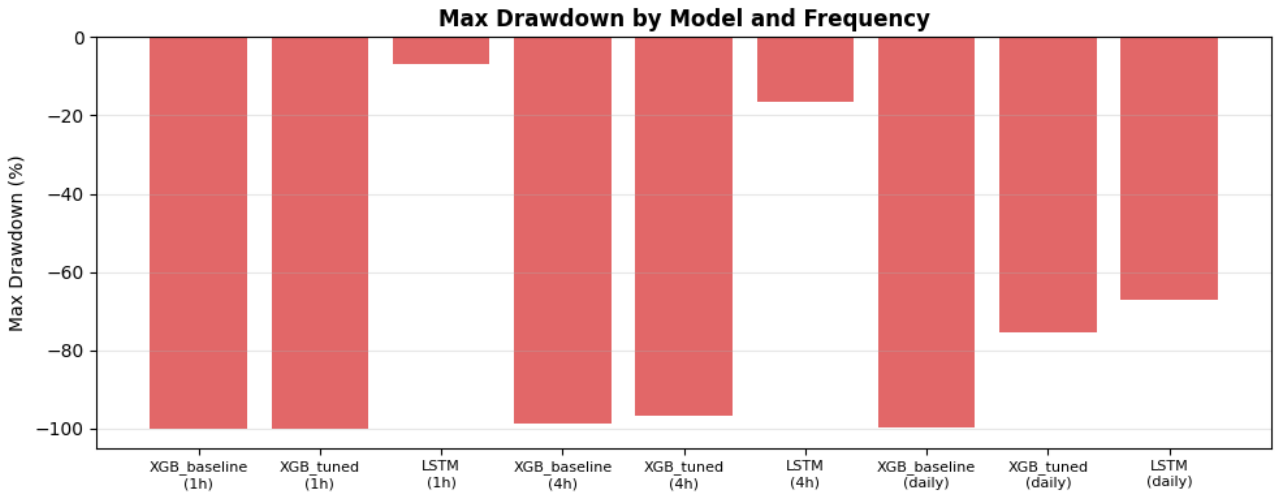
Cost is applied whenever the signal changes:
- Entry: signal 0 -> 1 (long): -0.1% applied
- Exit: signal 1 -> 0 (flat): -0.1% applied

Formula: strategy_return[t] = pct_return[t] * signal[t] - cost[t]
where cost[t] = |signal[t] - signal[t-1]| * 0.001

Impact by frequency:
- daily: fewer trades, lower total cost impact
- 4h: moderate trading frequency
- 1h: highest trading frequency, highest cost impact

Note: This model approximates CEX spot trading costs.
Real costs may include: spread, gas fees (for DEX), funding rates.
Gas fees on Ethereum can significantly increase costs for small positions.

Freq	Model	# Trades	Total Cost	Net Return
1h	XGB_baseline	2676	267.60%	-100.00%
1h	XGB_tuned	2716	271.60%	-100.00%
1h	LSTM	6	0.60%	-2.42%
4h	XGB_baseline	888	88.80%	-98.50%
4h	XGB_tuned	944	94.40%	-96.72%
4h	LSTM	16	1.60%	24.89%
daily	XGB_baseline	139	13.90%	-99.61%
daily	XGB_tuned	129	12.90%	-72.61%
daily	LSTM	22	2.20%	-31.27%



8. Conclusions

What worked in v2:
* Real data pipeline from Dune Analytics and The Graph proved more stable than synthetic data for training ML models on crypto price prediction.
* Multi-frequency approach allows capturing different market dynamics: daily signals capture macro trends, 4h captures medium-term moves, 1h captures short-term momentum.
* Fear & Greed Index adds useful sentiment signal not present in price data.
* price_divergence feature (abs(close - priceUSD)/priceUSD) captures CEX vs DEX price dislocations, a unique on-chain signal.
* Transaction cost modeling gives more realistic performance estimates.
* max_drawdown metric helps identify models with better risk profiles.

What did not work as expected:
* LSTM training is slow on hourly data (~43k rows) even with patience=15.
* On-chain holder data from Dune may have gaps for early periods.
* The Graph CSV coverage depends on data export date and may not cover 5 years.
* High-frequency models (1h) suffer more from transaction costs.

General observations:
* XGBoost typically outperforms LSTM on tabular financial data.
* Walk-forward validation shows significant variance across folds, suggesting regime changes in the crypto market over 5 years.
* The best performing frequency and model may change in different market conditions.

9. Next Steps

Recommended improvements:
* Ensemble: combine XGBoost and LSTM predictions with dynamic weighting.
* Dynamic threshold: use percentile-based threshold instead of fixed 0.5.
* Position sizing: implement Kelly criterion or volatility-based sizing.
* Add funding rates from perpetual futures as additional signal.
* Add social sentiment (Twitter/Reddit volume) as complementary signal.
* Explore transformer architectures (Informer, Temporal Fusion Transformer).
* Implement proper walk-forward backtesting with expanding or rolling windows.
* Add correlation analysis: UNI vs BTC, ETH, DeFi index.
* Regime detection: identify bull/bear market regimes explicitly.
* Gas cost adjustment: for DEX trading strategies, add gas fee model.

10. Limitations

Model limitations:
* Look-ahead bias risk: verify all features use only data available at time t.
* Survivorship bias: UNI survived and grew; selecting winners introduces bias.
* Execution assumption: strategy assumes execution at close price, no slippage.
* Market impact: model does not account for impact of large orders.
* Overfitting risk: with hundreds of features and complex models.

Data limitations:
* Dune Analytics data quality depends on the specific query and export date.
* The Graph may have incomplete data for very early protocol periods.
* Binance data may have gaps during maintenance periods.
* Fear & Greed Index is limited to ~1000 days of history.

Operational limitations:
* Model retraining required periodically as market regimes change.
* API rate limits may cause incomplete data downloads.
* LSTM training time scales poorly with data size.

Disclaimer: This report is for educational purposes only.
It does not constitute investment advice.

11. Errors During Execution

No errors encountered during pipeline execution.